

Intelligent Candidate Discovery & Ranking

Ranking 100,000 candidates for a Senior AI Engineer role —
by understanding fit, not matching keywords.

Team TODO — your team name

Lead TODO — team leader

Redrob Data & AI Challenge

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Solution Overview

A hybrid ranker — transparent rule layer + local semantic embeddings — scoring all 100K candidates in ~1.6s on CPU, offline.

Evidence over keywords

Backbone signal reads career-history descriptions (what they built) — 'vector DB' is in 12,866 skill lists but only 108 descriptions.

Role-class gate

A 'Marketing Manager with 11 AI skills' keeps only ~39% of its score — structurally can't reach the top.

Availability matters

206 days idle + 7% response = not hireable. Behavioral signals are a first-class multiplier.

Impossible profiles gated

Five consistency checks exclude the ~80 honeypots before they reach the shortlist.

JD Understanding & Candidate Evaluation

What the role actually needs

- Production embeddings retrieval + vector DB / hybrid search + learning-to-rank / recsys, at product companies.
- 5–9 yrs (ideal 6–8); has shipped a search/ranking/recsys system at scale; NDCG/MRR/A-B eval rigor.
- Pune/Noida or willing to relocate; sub-30-day notice preferred; no visa sponsorship outside India.

Traps it deliberately rejects

- Keyword stuffers — non-engineers with AI skills listed; self-learner 'side-project' RAG; recent-only LangChain.
- Pure-consulting careers (TCS/Infosys/Wipro...); CV/speech/robotics without NLP/IR; research-only without production.
- Our answer: score the gap between what the JD says and means — production evidence >> title >> skills >> semantic fit.

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Ranking Methodology

```
content = 0.46·PES + 0.10·EMB + 0.14·Title + 0.10·Skills + 0.08·Exp + 0.04·Loc + 0.08·Base
```

```
final    = (content + synergy) · RoleClassGate · TrapPenalty · Availability · HoneypotGate
```

PES — Production-Evidence	0.46	retrieval/ranking/recsys/RAG/eval in career descriptions × company × duration × recency
EMB — semantic	0.10	bge-small cosine vs JD-intent; calibrated so genuine fits sit above pool p99
Title + role-class gate	0.14	core AI title ×1.0; non-tech ×0.06 — the anti-keyword-stuffer lever
Skills (corroborated)	0.10	endorsements+duration+proficiency; expert-at-0-months = red flag
Behavioral availability	×	response, recency, notice, open-to-work → multiplier [0.80, 1.12]
Trap / honeypot gates	×	demote stuffer/self-learner/consulting/CV-only; exclude impossible profiles

Additive for graded factors; multiplicative for gates — one disqualifier can't be out-voted by keyword volume.

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Explainability & Data Validation

Explainable & grounded

- Fact-grounded reason per rank, no LLM: years, title, company, evidence, skills.
- Up to two honest concerns disclosed.
- Assertion: every named skill/employer exists — 0 hallucinations.

5 impossibility checks

- ≥ 3 expert skills with 0 months used
- role tenure > what its dates allow
- career sum > stated experience + 3y
- summary tenure vs field off by > 4y
- ≥ 4 skills used longer than the career

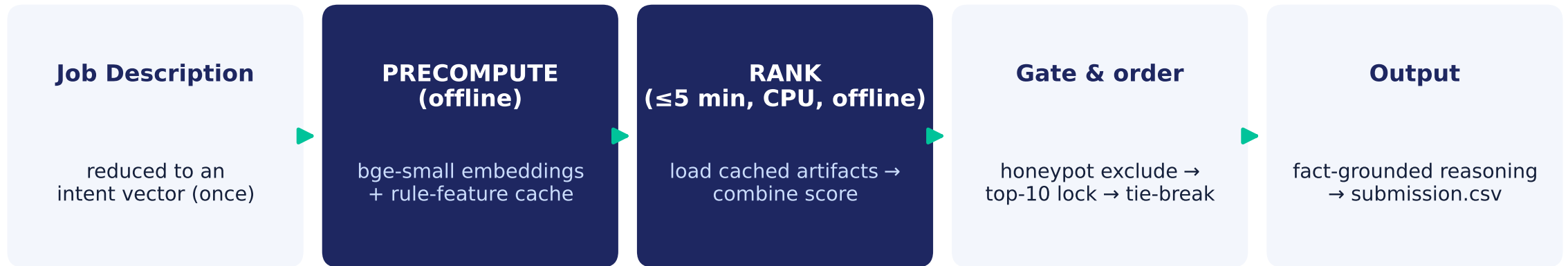
Deliberately ignored as synthetic noise

Salary min > max (~19% of pool) and any single skill-duration overrun (~9%) are pervasive in the data — using them as honeypot signals would bury thousands of genuine candidates. We use only rare, false-positive-free checks.

Result: the ~80 honeypots are flagged and excluded; 0 reach the top-100.

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End-to-End Workflow



The expensive embedding work is one-time and offline; the judged ranking step only reads cached artifacts — so it is CPU-only, network-free, and finishes in ~1.6 seconds.

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System Architecture

OFFLINE · precompute.py (network + time allowed)

bge-small-en-v1.5

JD + candidate → cand_embeddings.npy

rule features

PES · title · skills · behavioral · traps → features.npz

ONLINE · rank.py (CPU-only · no network · ~1.6s)

features

signals + honeypot/trap

scoring

weights · top-10 lock · valid emit

reasoning

grounded NL, no LLM

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Results & Performance

100/100

top-100 are genuine
AI/ML/NLP/DS roles

0

honeypots & traps
in the top-100

~1.6 s

ranking step
(limit: 5 min)

90/100

India-based;
all product/AI/big-tech

Ranking quality

- 82 honeypots flagged pool-wide (of ~80 seeded); 0 in the top-100 — far under the 10% disqualification line.
- Behavioral down-weighting verified: a PES-0.98 but 206-day-idle / 7%-response candidate correctly sinks to the tail.
- Reasoning: 0 hallucinations, all distinct, tone matches rank. Output passes the official format validator.

Constraints met: ≤ 5 min $\rightarrow$ ~1.6s · ≤ 16 GB $\rightarrow$ <1 GB · CPU-only ✓ · no network during ranking ✓

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Technologies Used

Python · numpy

Vectorized scoring + a single matmul for cosine over 100K vectors.

sentence-transformers · PyTorch (CPU)

BAAI/bge-small-en-v1.5 — small, fast, strong retrieval embeddings; used offline only.

No vector DB / FAISS at rank time

One query over 100K vectors is one matmul; an ANN index adds dependency + non-determinism for no gain.

Deterministic, explainable heuristics

No black-box LTR training (no labels); every weight is auditable for the code-reproduction & interview.

Chosen to maximize quality within a hard CPU / offline / 5-minute budget — and to stay defensible.

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Submission Assets

GitHub repo

`github.com/<you>/redrob-ranker`

Reproduce

```
python rank.py --candidates ./candidates.jsonl --out ./submission.csv
```

Sandbox / demo

`huggingface.co/spaces/<you>/redrob-ranker`

Ranked output

`submission.csv` (top-100)

Walkthrough video

`<walkthrough video link>`

Build something real. Make hiring smarter.